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**Hard

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The control logic is implemented physically using digital hardware circuits (logic gates, flip-flops, decoders).

control

The control signals are generated by executing a tiny, low-level program (a **Microprogram**) stored in a special, fast memory called **Control Memory (ROM)**.

microin

Feature	Hardwired CU	Microprogrammed CU
Control Logic	Fixed, implemented using logic gates and flip-flops.	Programmable, implemented using microprograms stored in control memory.
Flexibility	Low; changing instructions requires physical changes to the hardware.	High; new instructions can be added by loading new microprograms.
Performance	Generally faster due to direct hardware implementation.	Slower due to the overhead of fetching and executing microprograms.
Cost	Higher initial cost due to complex hardware.	Lower initial cost due to simpler hardware and reprogrammability.
Scalability	Difficult to scale for complex instruction sets.	Easier to scale by adding more microprograms.
Reliability	High; fewer components and no moving parts.	Lower; more components and potential for microprogram errors.
Development	Complex and time-consuming to design.	Easier and faster to develop using microprogramming techniques.

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 $\gamma > 1$. **

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> **Quick Recall**

> `Con
Slower